

Fundamental Ground Experiment of a Water Resistojet Propulsion System: AQUARIUS Installed on a 6U CubeSat: EQUULEUS

By Jun ASAKAWA,¹⁾ Hiroyuki KOIZUMI,^{1),2)} Keita NISHII,²⁾ Naoki TAKEDA,²⁾ Masaya MUROHARA,¹⁾
 Ryu FUNASE,¹⁾ and Kimiya KOMURASAKI¹⁾

¹⁾ Department of Aeronautics and Astronautics, The University of Tokyo, Tokyo, Japan

²⁾ Graduate School of Frontier Sciences, The University of Tokyo, Tokyo, Japan

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In this study, a water resistojet propulsion system, named AQUARIUS, is proposed and the breadboard model (BBM) test results and the engineering model (EM) design are described. AQUARIUS will be on-boarded the SLS EM-1 CubeSat: EQUULEUS, which is scheduled to be launch in 2019. AQUARIUS uses a storable, safe and non-toxic propellant: water, which allows for the design of all propulsion system below 100 kPa, reduction of dry mass ratio and simplification of feed line routing using soft tubes. AQUARIUS consists of three components as follows: a tank, a vaporization chamber and thruster head. BBM of the vaporization chamber, and the thruster head were designed and tested. The evaporation rate required heat and thrust performance were evaluated. Based on the BBM test results, the engineering model of AQUARIUS was designed.

Key Words: Resistojet, Water, Propulsion, CubeSat, Nanosatellite

Nomenclature

A	: area
c^*	: characteristic velocity
F	: thrust
g	: gravitational acceleration
I_{sp}	: specific impulse
$\dot{m}$	: mass flow rate
M	: molecular weight
p	: pressure
R	: gas constant
T	: temperature
v	: thermal velocity
α	: evaporation coefficient

Subscripts

g	: gas
l	: liquid
t	: throat
v	: vapor

1. Introduction

The utilization of micro/nano-satellites has been expanded drastically since 2010.¹⁾ Along with that, various micro/nano-propulsion systems have been proposed, designed and demonstrated.²⁾ Micro-solid motor has a high capability with micro/nano-satellites from the viewpoint of simplicity.^{3,4)} In 2014, the world's first interplanetary micro-spacecraft: PROCYON was developed and launched by The University of Tokyo and JAXA.⁵⁾ Ion thruster and cold gas thruster unified propulsion system (I-COUPS) was on-boarded PROCYON and operated more than 200 hours in deep space.^{6,7)} In addition,

deep space exploration missions by using 6U CubeSat have been planned in recent years.⁸⁻¹⁰⁾

Micro-propulsion systems are indispensable for micro/nano-satellites to conduct such advanced missions. However, there are some problems such as low thrust performance or higher dry mass fraction because of high-pressure gas systems such as a tank, feed lines, and valves. For example, in the case of miniature ion propulsion system "MIPS" developed by the University of Tokyo, 45 % of the total propulsion system with a mass of 10 kg was occupied with high-pressure gas systems which did not contribute to thrust performance directly.¹¹⁾ Chemical propulsion systems which use hydrogen or ADN are also conventional for large satellites. However, utilizing cost of toxic propellant including hardware, operation, ground support infrastructure and transportation will reach to more than \$1M.¹²⁾ It is difficult for the solid motor to meet safety requirements for secondary payload. Therefore, these are not feasible for micro/nano-satellites. Requirements for the next-generation micro/nano-propulsion system to expand the use of micro/nano-satellites are as follows:

- Safety availability and easy-handling for a piggy-back launch and a short-span development.
- Non high-pressure gas system for volume and weight saving.
- Multi functions such as delta-V maneuver or reaction control system.

AQUA resistojet propulsion system (AQUARIUS) has been developed since April. 2016.¹³⁾ AQUARIUS uses a storable, safe and non-toxic propellant: water. Liquid propellant storage allows for the design of all propulsion system below 100 kPa. What is the most difficult is high latent heat of water. To overcome this difficulty, waste heat generated from communication components are reused to vaporize water.

AQUARIUS will be on-boarded the interplanetary nanospacecraft: EQUULEUS, which has been developed by The University of Tokyo and JAXA.^{14,15} EQUULEUS is scheduled to be launched in 2019 by the Space Launch System (SLS). The delta-V maneuver and the reaction control system are required to AQUARIUS. Thrust performance specifications of AQUARIUS are listed in Table 1. In this paper, breadboard model (BBM) test results and engineering model (EM) design are described.

Table 1. Specifications of AQUARIUS.

AQUARIUS	
Propellant	Water (< 100 kPa)
Thrust	4.0 mN for Delta-V thruster 2.0 mN for RCS thruster
Specific impulse	70 s
Heating	+ 100 K
Power consumption	< 20 W
Volume (CubeSat unit)	2.0 U

2. AQUARIUS

Thrust generating process of AQUARIUS is simple. At first, water is vaporized at saturated vapor pressure and a room temperature. Then saturated water vapor is heated and accelerated at a thruster head. AQUARIUS consists of a tank, a vaporization chamber, pre-heaters, and nozzles. Figure 1 shows a system diagram of the AQUARIUS. AQUARIUS has two delta-V thruster heads and four RCS thruster heads. Water is stored in a bladder inserted in the tank.

Pressurized gas under 100 kPa is stored in the gap space between the tank and the bladder. The water is supplied from the bladder to the vaporization chamber by opening a valve for a blowdown mode. The bladder has been already installed on a monopropellant propulsion system using hydrogen peroxide and demonstrated in an orbit.¹⁶⁾ After water supply, the vaporization chamber is filled with saturated water vapor. The vaporization chamber is heated by electrical heaters and waste heat from communication components. Saturated water vapor is supplied to the pre-heater and heated. The pre-heater, made of aluminum, has a helical flow pass in it to increase heating area. Water vapor which flows in the helical flow path is heated by heating the outside of aluminum. Heated water vapor is accelerated at the nozzle. We have been testing a water feeding system and measuring thrust.

3. Setup and Method of the BBM Test

3.1. Vaporization chamber

BBM of the vaporization chamber was designed and tested. Details of the test are described below.

3.1.1. Experimental setup

The experimental setup is shown in Fig. 2a. All experiments were conducted in a vacuum chamber with a diameter of 1.0 m and a length of 1.7 m. The diameter of the vaporization chamber was 40 mm and the depth was 30 mm. One side of the vaporization chamber was sealed by a glass flange to observe inside. Water was stored in the bladder which was located

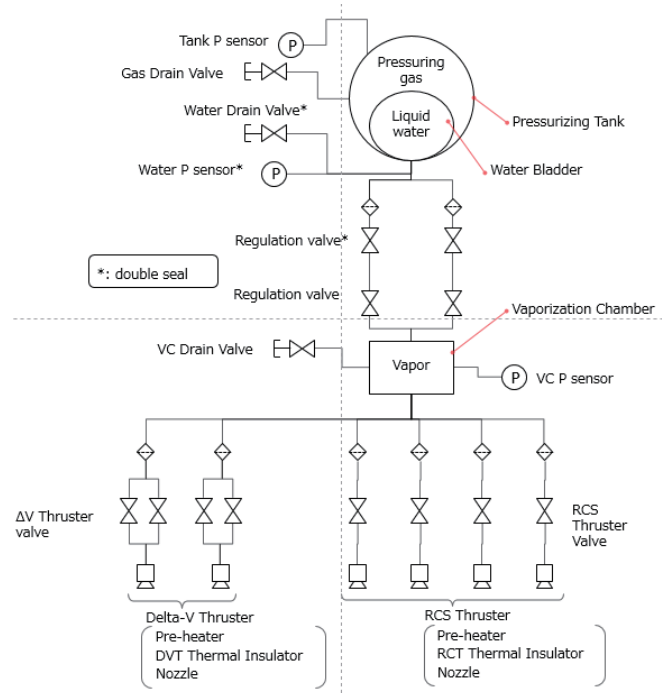


Fig. 1. A system diagram of AQUARIUS.

upstream of the vaporization chamber. The vaporization chamber and the bladder were connected through a solenoid valve. Water stored in the bladder was injected into the vaporization chamber by actuating the solenoid valve. The vaporization chamber was mounted on a temperature control board which was connected to a chiller. The valve tip and the water injection surface were heated by using electrical heaters. A conical nozzle with a throat diameter of 1.2 mm and an expansion ratio of 4.0 was integrated downstream of the vaporization chamber. A Pressure sensor was mounted on the side wall of the vaporization chamber. Thermocouples were mounted on the inner wall and the water injection surface.

3.1.2. Experimental method

The vaporization chamber pressure is dominated by the saturated vapor pressure depending on the droplet temperature. By using Hertz-Knudsen equation, the evaporation rate is expressed as

$$\dot{m} = \alpha \left(p_v \sqrt{\frac{M}{2\pi RT_l}} - p_g \sqrt{\frac{M}{2\pi RT_g}} \right) A_l = \frac{2\alpha}{\pi} \left(\frac{p_v}{\bar{v}_g} - \frac{p_g}{\bar{v}_g} \right) A_l. \quad (1)$$

During the operation, the evaporation rate is in balance with the choke mass flow rate at the nozzle throat. Therefore, the balance equation is expressed as

$$\frac{2\alpha}{\pi} \left(\frac{p_v}{\bar{v}_g} - \frac{p_g}{\bar{v}_g} \right) A_l = \frac{A_t p_g}{c^*}. \quad (2)$$

In this study, the vaporization chamber pressure history, temperature history and a behavior of droplet were investigated while changing the valve open duration time and the heater

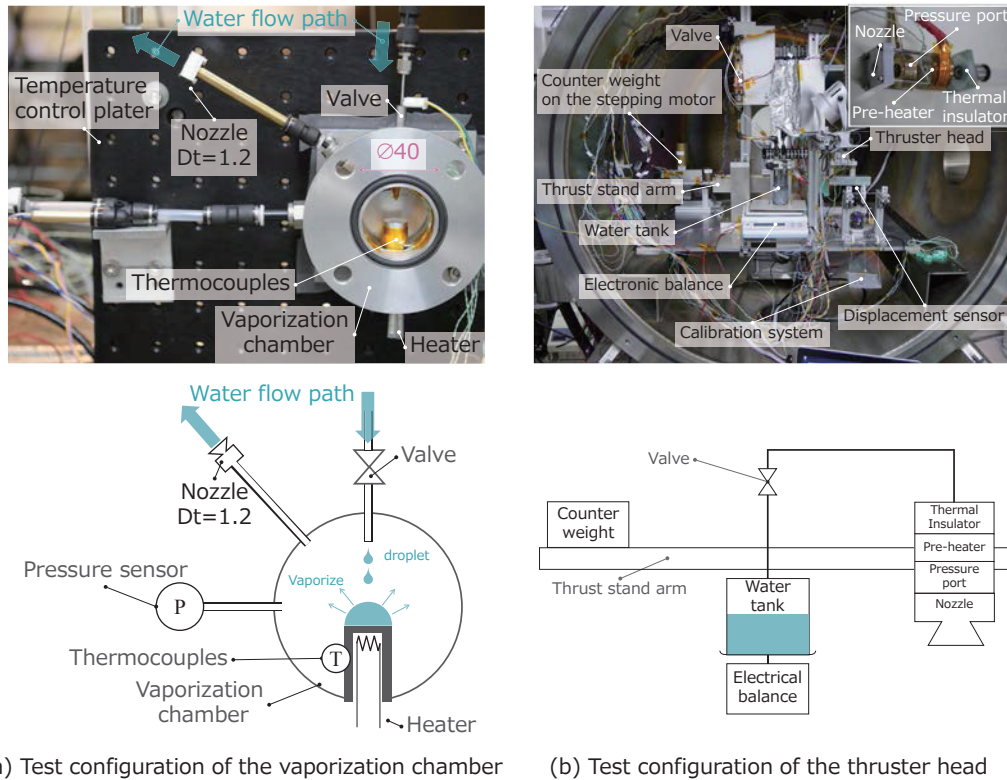


Fig. 2. Experimental setup of the BBM test.

input power. The valve open frequency is arranged depending on the valve open duration time.

3.2. Thruster head

BBM of the thruster head was designed, and its performance was evaluated by measuring thrust in a vacuum. Details of the test are described as follows.

3.2.1. Experimental setup

The experimental setup is shown in Fig. 2b. The thruster head was composed of four parts as follows: a thermal insulator, a pre-heater, a pressure port and a nozzle. The thermal insulator, which was made of glass epoxy, had a role of insulating the pre-heater and upstream of the thrust head. The pre-heater, which was made of aluminum, was manufactured by using 3D printing technology. A heater tape was wrapped around the cylindrical pre-heater. A couple types of the nozzle, which had different throat diameter, expansion angle, and expansion ratio, were evaluated. The thruster was measured by using a torsional thrust stand in the vacuum chamber which was similar to the vaporization chamber test. The thruster head was mounted on the thrust stand arm. The water tank was mounted on an electrical balance located on the outside of the thrust stand to measure a mass flow rate. Saturated water vapor stored in the tank was supplied to the thruster head by opening the valve. Gravity center point of the thrust stand arm was arranged by counter-weights. The displacement generated by the thrust was measured by using an inductive displacement sensor. Thrust measurement sensitivity was $10 \mu\text{N}$. Propellant feeding tubes and electrical cables were coiled to suppress the influence of rigidity or friction of them.

3.2.2. Experimental method

Thrust and pressure history were investigated while

changing the nozzle design, pre-heater design and tank temperature. The thrust measurement system was calibrated before and after all tests. Mass flow rate of the water was also measured by using the electrical balance. Specific impulse was expressed as

$$I_{sp} = \frac{F}{\dot{m}g}. \quad (3)$$

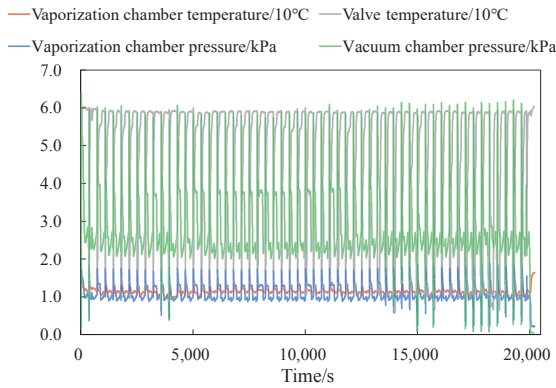
4. BBM Experimental Result

4.1. Vaporization chamber

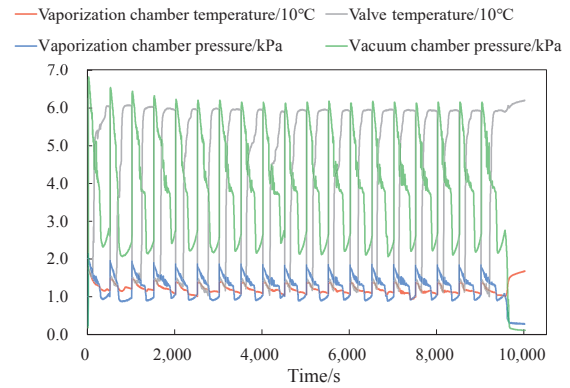
Some test results are shown in Fig. 3. The vaporization chamber operated stably for more than a couple of hours without freezing. Just after the injection, liquid water was supplied and the vaporization chamber pressure increased sharply. The vaporization chamber pressure decreased gradually because heat transferred from the vaporization chamber to water. The vaporization chamber pressure became constant when the evaporation rate is in balance with the choke mass flow rate.

4.2. Thruster head

Test results of thruster head for RCS thruster are shown in Fig. 4. The pre-heater was kept at about 100°C . The thruster head operated stably at a thrust of 1.6 mN at the pressure before nozzle of 1.0 kPa although designed thrust was 2.0 mN for RCS. This is because water flow path between tank and thruster head of the ground test configuration is longer than that of the actual satellite configuration, leading to the increment of pressure loss. If the pressure before nozzle is kept more than 1.0 kPa , a thrust

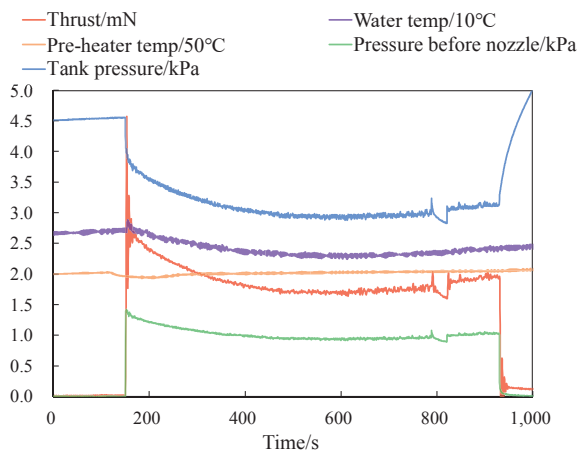


(a) Test result (valve open time: 0.50 s, heater: 10 W)

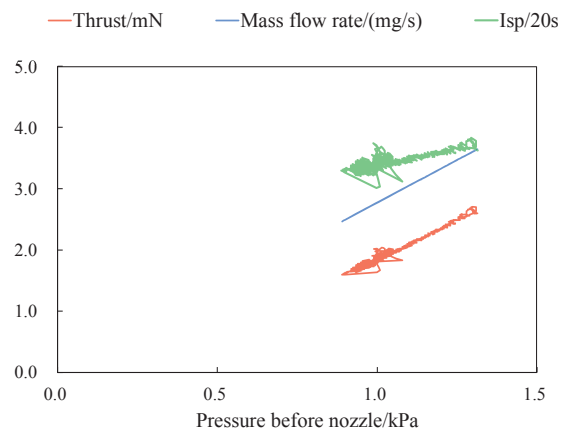


(b) Test result (valve open time: 1.0 s, heater: 10 W)

Fig. 3. Test results of the BBM vaporization chamber.



(a) Time history of the thruster head test



(b) Specific Impulse vs Pressure before nozzle

Fig. 4. Test results of the BBM thruster-head.

of 2.0 mN and a specific impulse of 70 s can be achieved.

5. EM Design

Engineering model (EM) was designed based on the BBM test results. Details are described below.

5.1. Tank and vaporization chamber

EM of the tank and vaporization chamber are shown in Fig. 5a. The bladder is inserted in the tank. Inlet of the tank is sealed by the tank attachment. Drain valves for propellant, pressurized gas (< 60 kPa nominal) and purge gas for the vaporization chamber are connected. The tank and the vaporization chamber are connected through the regulation valves and the relay block. The vaporization chamber and the relay block, which were made of aluminum, were manufactured by using the 3D printing technology to make flow paths inside of them. Eight thruster valves are inserted in the vaporization chamber. The vaporization chamber and the thruster heads are connected through the joints and soft tubes. Dry mass of the combination

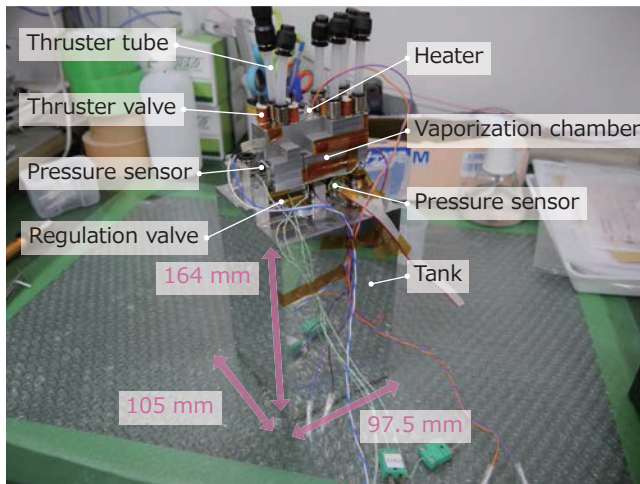
of the tank and the vaporization chamber is 1.2 kg.

5.2. Thruster head

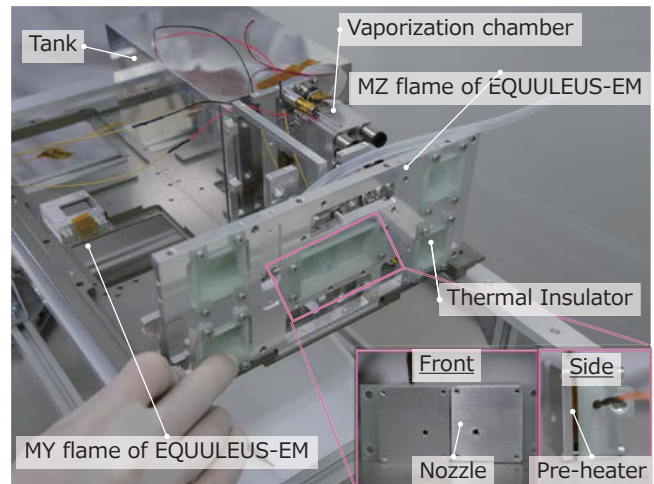
EM of the thruster head is shown in Fig. 5b. The pre-heater, which is made of aluminum, is manufactured by using 3D printing technology. Flow path diameter of the pre-heater is 4 mm, and the total flow path length of it is about 130 mm. Water vapor is heated by the electrical heater wrapped outside the pre-heater. Throat diameter of the nozzle for reaction control is 1.2 mm, and that of the nozzle for delta-V maneuver is 2.6 mm.

6. Conclusion

- Water resistojet propulsion system: AQUARIUS for a CubeSat was proposed.
- Fundamental ground experiments using breadboard model were conducted while changing the experimental conditions or shapes, and successfully finished.
- Engineering model of AQUARIUS was designed and its status was indicated.



(a) AQUARIUS-EM



(b) AQUARIUS-EM integrated on EQUULEUS-EM

Fig. 5. Engineering model of AQUARIUS.

Acknowledgments

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